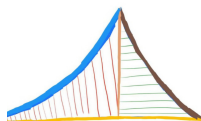


Matrices In Geometry

1



G. V. V. Sharma

ABOUT THIS BOOK

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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First manual appeared in January 2018

First edition published on July 10, 2024

iIn this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

CONTENTS

1 **GATE-BM-2025**

4

1. GATE-BM-2025

Math for EE

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1) GATE BM 2025

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1.0.1 The 12 musical notes are given as $C, C^\#, D, D^\#, E, F, F^\#, G, G^\#, A, A^\#, B$. Frequency of each note is $\sqrt[12]{2}$ times the frequency of the previous note. If the frequency of the note C is 130.8 Hz, then the ratio of frequencies of notes $F^\#$ and C is: (GATE BM 2025)

- (a) $\sqrt[6]{2}$
- (b) $\sqrt{2}$
- (c) $\sqrt[4]{2}$
- (d) 2

Solution:

Each successive musical note has a frequency $\sqrt[12]{2}$ times that of the previous note. From C to $F^\#$, there are 6 successive intervals:

$$C \rightarrow C^\# \rightarrow D \rightarrow D^\# \rightarrow E \rightarrow F \rightarrow F^\# \quad (1.0.1.1)$$

Therefore,

$$\frac{f_{F^\#}}{f_C} = \left(\sqrt[12]{2} \right)^6 \quad (1.0.1.2)$$

$$= 2^{6/12} = 2^{1/2} = \sqrt{2} \quad (1.0.1.3)$$

Hence,

$$\boxed{\frac{f_{F^\#}}{f_C} = \sqrt{2}} \quad (1.0.1.4)$$

$$(b) \sqrt{2} \quad (1.0.1.5)$$

codes/bm1.py

1.0.2 The following figures show three curves generated using an iterative algorithm. The total length of the curve generated after “Iteration n ” is:

GATE BM 2025

1

Iteration 0:

Length of segment: 1

$$L_n = \left(\frac{5}{3}\right)^n \quad (1.0.2.5)$$

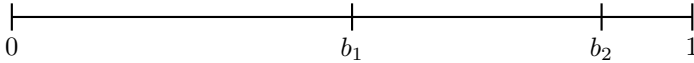
Hence, the correct answer is

$$(b) \left(\frac{5}{3}\right)^n \quad (1.0.2.6)$$

1.0.3 A stick of length one meter is broken at two locations at distances of b_1 and b_2 from the origin (0), as shown in the figure. Note that $0 < b_1 < b_2 < 1$. Which one of the following is NOT a necessary condition for forming a triangle using the three pieces?

Note: All lengths are in meter. The figure shown is representative.

(GATE BM 2025)



- (A) $b_1 < 0.5$
- (B) $b_2 > 0.5$
- (C) $b_2 < b_1 + 0.5$
- (D) $b_1 + b_2 < 1$

Solution:

The stick of length 1 m is broken at b_1 and b_2 .

Therefore, the lengths of the three pieces are

$$l_1 = b_1, \quad l_2 = b_2 - b_1, \quad l_3 = 1 - b_2. \quad (1.0.3.1)$$

For three lengths to form a triangle, the sum of any two sides must be greater than the third side.

Therefore,

$$l_1 + l_2 > l_3 \quad (1.0.3.2)$$

$$b_1 + (b_2 - b_1) > 1 - b_2 \quad (1.0.3.3)$$

$$b_2 > 1 - b_2 \quad (1.0.3.4)$$

$$b_2 > \frac{1}{2}. \quad (1.0.3.5)$$

Hence, option (B) is a necessary condition.

Now consider

$$l_1 + l_3 > l_2. \quad (1.0.3.6)$$

Substituting the lengths,

$$b_1 + (1 - b_2) > b_2 - b_1. \quad (1.0.3.7)$$

Therefore,

$$2b_1 + 1 > 2b_2 \quad (1.0.3.8)$$

or

$$b_2 < b_1 + \frac{1}{2}. \quad (1.0.3.9)$$

Hence, option (C) is also a necessary condition.

Finally,

$$l_2 + l_3 > l_1. \quad (1.0.3.10)$$

Substituting,

$$(b_2 - b_1) + (1 - b_2) > b_1. \quad (1.0.3.11)$$

Therefore,

$$1 - b_1 > b_1 \quad (1.0.3.12)$$

which gives

$$b_1 < \frac{1}{2}. \quad (1.0.3.13)$$

Hence, option (A) is also a necessary condition.

However, the condition

$$b_1 + b_2 < 1 \quad (1.0.3.14)$$

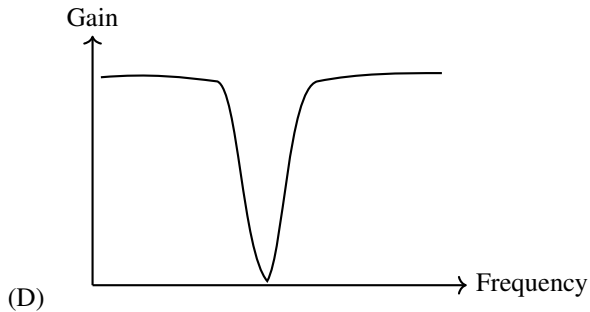
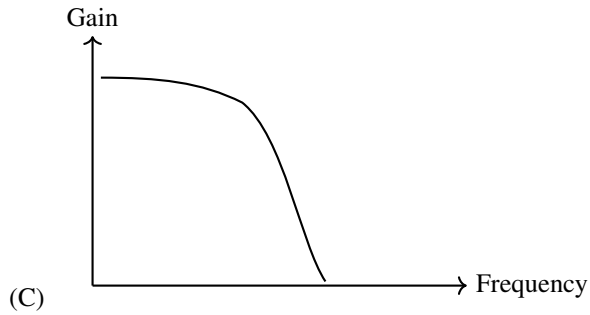
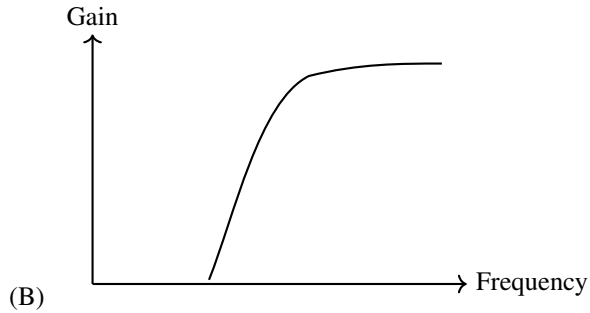
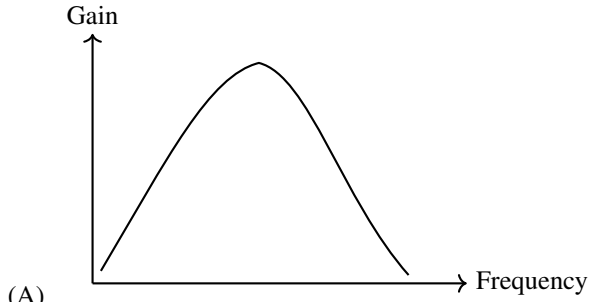
is not required for the three pieces to form a triangle.

Therefore, the condition which is **NOT** necessary is (D)

codes/bm3.py

1.0.4 Which one of the following represents the frequency response of a notch filter?

(GATE BM 2025)



Solution:

A notch filter is a band-stop filter that rejects or strongly attenuates a narrow range of frequencies while allowing frequencies outside that range to pass.

Therefore, its frequency response has high gain at low and high frequencies, with a sharp reduction in gain at a particular frequency.

Mathematically, if ω_0 is the rejected frequency, then the response has a deep attenuation around ω_0 :

$$|H(j\omega_0)| \approx 0 \quad (1.0.4.1)$$

while frequencies away from ω_0 are passed with relatively high gain.

$$|H(j\omega)| \text{ is high for } \omega < \omega_0 \text{ and } \omega > \omega_0. \quad (1.0.4.2)$$

Among the given frequency responses, only option (D) shows a sharp reduction in gain at a particular frequency while allowing frequencies on either side to pass. Hence, Answer: (D)

- 1.0.5 For any continuous and differentiable function $f(x)$ with first derivative $f'(x) \neq 0$ for all values of x , which one of the following is ALWAYS TRUE for $a \neq b$? Here, a and b are finite and real.

(GATE BM 2025)

- (A) $f(a) > f(b)$
- (B) $f(a) = f(b)$
- (C) $f(a) \neq f(b)$
- (D) $f(a) < f(b)$

Solution:

We are given that $f(x)$ is continuous and differentiable, and

$$f'(x) \neq 0 \quad \text{for all } x. \quad (1.0.5.1)$$

Suppose, for contradiction, that

$$f(a) = f(b). \quad (1.0.5.2)$$

Since $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b) , Rolle's theorem can be applied.

Rolle's theorem states that if

$$f(a) = f(b), \quad (1.0.5.3)$$

then there must exist at least one point $c \in (a, b)$ such that

$$f'(c) = 0. \quad (1.0.5.4)$$

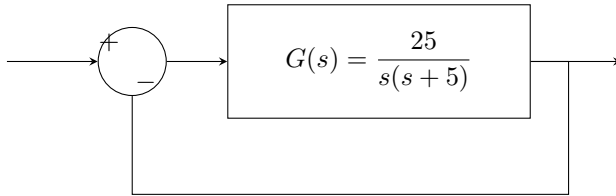
But this contradicts the given condition

$$f'(x) \neq 0 \quad \text{for all } x. \quad (1.0.5.5)$$

Therefore, our assumption that $f(a) = f(b)$ is false. Hence,

$$\boxed{f(a) \neq f(b)}. \quad (1.0.5.6)$$

1.0.6 The closed loop system shown below _____.



(GATE BM 2025)

- (A) is overdamped
- (B) is underdamped
- (C) is critically damped
- (D) has sustained oscillations

Solution:

The forward transfer function is:

$$G(s) = \frac{25}{s(s + 5)}$$

Using the Laplace transform feedback relation for a unity negative feedback system, the closed-loop transfer function is:

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$

Substitute $G(s)$:

$$\frac{C(s)}{R(s)} = \frac{\frac{25}{s(s+5)}}{1 + \frac{25}{s(s+5)}} = \frac{25}{s^2 + 5s + 25}$$

The characteristic equation in the Laplace domain is:

$$s^2 + 5s + 25 = 0$$

Comparing this with the standard second-order system equation $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$:

$$\omega_n^2 = 25 \implies \omega_n = 5$$

$$2\zeta\omega_n = 5 \implies 2\zeta(5) = 5 \implies \zeta = 0.5$$

Since $0 < \zeta < 1$, the system is underdamped.

To find the impulse response in the time domain, we compute the Inverse Laplace Transform of $C(s)$ (for $R(s) = 1$):

$$C(s) = \frac{25}{s^2 + 5s + 25}$$

Completing the square in the denominator:

$$s^2 + 5s + 25 = \left(s + \frac{5}{2}\right)^2 + \frac{75}{4} = (s + 2.5)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2$$

Rewriting $C(s)$:

$$C(s) = \frac{25}{\left(s + \frac{5}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2} = \frac{25}{\frac{5\sqrt{3}}{2}} \cdot \frac{\frac{5\sqrt{3}}{2}}{\left(s + \frac{5}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2}$$

$$C(s) = \frac{10}{\sqrt{3}} \cdot \frac{\frac{5\sqrt{3}}{2}}{\left(s + \frac{5}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2}$$

Taking the Inverse Laplace Transform $\mathcal{L}^{-1}\{C(s)\}$:

$$c(t) = \frac{10}{\sqrt{3}} e^{-\frac{5}{2}t} \sin\left(\frac{5\sqrt{3}}{2}t\right) \quad \text{for } t \geq 0$$

The presence of the decaying exponential $e^{-\frac{5}{2}t}$ multiplied by the sine term confirms an underdamped oscillatory response.

Correct option: (B) is underdamped

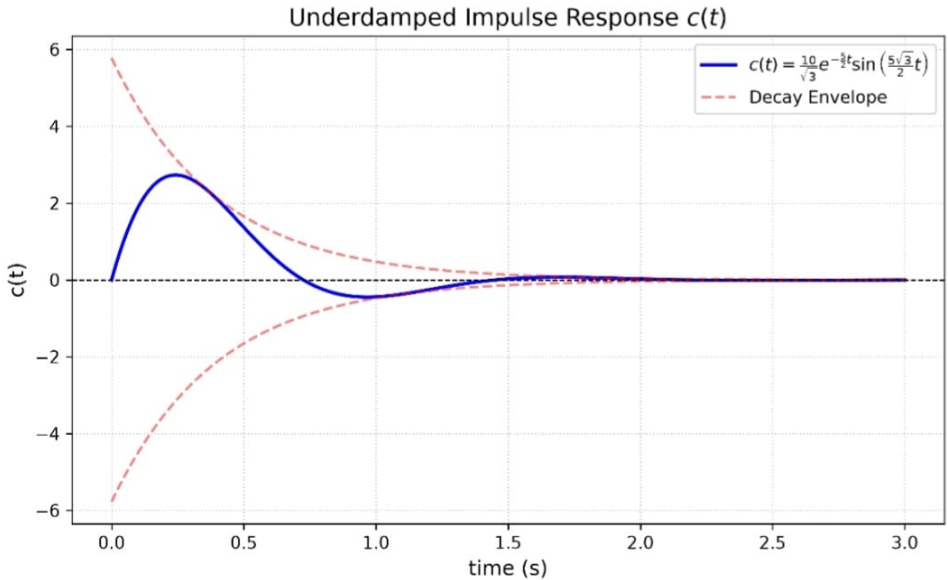


Fig. 1.0.6.1. $c(t)$ vs time

codes/bm6.py

1.0.7 The eigenvalues of the matrix

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad (1.0.7.1)$$

are _____.

(GATE BM 2025)

- (A) $\cos \theta$ and $\sin \theta$
- (B) $e^{i\theta}$ and $e^{-i\theta}$, where $i = \sqrt{-1}$
- (C) $\cos \theta$ and $-\sin \theta$
- (D) $\cos 2\theta$ and $\sin 2\theta$

Solution:

Let the given matrix be

$$A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}. \quad (1.0.7.2)$$

The eigenvalues λ are obtained from the characteristic equation

$$\det(A - \lambda I) = 0. \quad (1.0.7.3)$$

Therefore,

$$\begin{vmatrix} \cos \theta - \lambda & -\sin \theta \\ \sin \theta & \cos \theta - \lambda \end{vmatrix} = 0. \quad (1.0.7.4)$$

Expanding the determinant,

$$(\cos \theta - \lambda)^2 + \sin^2 \theta = 0. \quad (1.0.7.5)$$

Expanding further,

$$\cos^2 \theta - 2\lambda \cos \theta + \lambda^2 + \sin^2 \theta = 0. \quad (1.0.7.6)$$

Using

$$\sin^2 \theta + \cos^2 \theta = 1, \quad (1.0.7.7)$$

we get

$$\lambda^2 - 2\lambda \cos \theta + 1 = 0. \quad (1.0.7.8)$$

Using the quadratic formula,

$$\lambda = \frac{2 \cos \theta \pm \sqrt{4 \cos^2 \theta - 4}}{2}. \quad (1.0.7.9)$$

Since

$$4 \cos^2 \theta - 4 = -4 \sin^2 \theta, \quad (1.0.7.10)$$

we have

$$\lambda = \frac{2 \cos \theta \pm 2i \sin \theta}{2}. \quad (1.0.7.11)$$

Hence,

$$\lambda = \cos \theta \pm i \sin \theta. \quad (1.0.7.12)$$

Using Euler's formula,

$$e^{i\theta} = \cos \theta + i \sin \theta, \quad (1.0.7.13)$$

and

$$e^{-i\theta} = \cos \theta - i \sin \theta, \quad (1.0.7.14)$$

the eigenvalues are

$$\boxed{\lambda_1 = e^{i\theta}, \quad \lambda_2 = e^{-i\theta}}. \quad (1.0.7.15)$$

Therefore, the correct answer is (B) (1.0.7.16)

codes/bm7.py

1.0.8 If $f(x) = x - \frac{1}{x}$, the value of $\lim_{\Delta x \rightarrow 0} \frac{f(1+\Delta x) - f(1)}{\Delta x}$ is _____.
(GATE BM 2025)

- (A) 0
- (B) $\frac{1}{2}$
- (C) 1
- (D) 2

Solution:

The given limit is the definition of the derivative of $f(x)$ at $x = 1$:

$$f'(1) = \lim_{\Delta x \rightarrow 0} \frac{f(1 + \Delta x) - f(1)}{\Delta x}. \quad (1.0.8.1)$$

Given

$$f(x) = x - \frac{1}{x}, \quad (1.0.8.2)$$

differentiate with respect to x :

$$f'(x) = 1 + \frac{1}{x^2}. \quad (1.0.8.3)$$

Therefore,

$$f'(1) = 1 + \frac{1}{1^2} = 2. \quad (1.0.8.4)$$

Therefore, the correct option is (D) 2

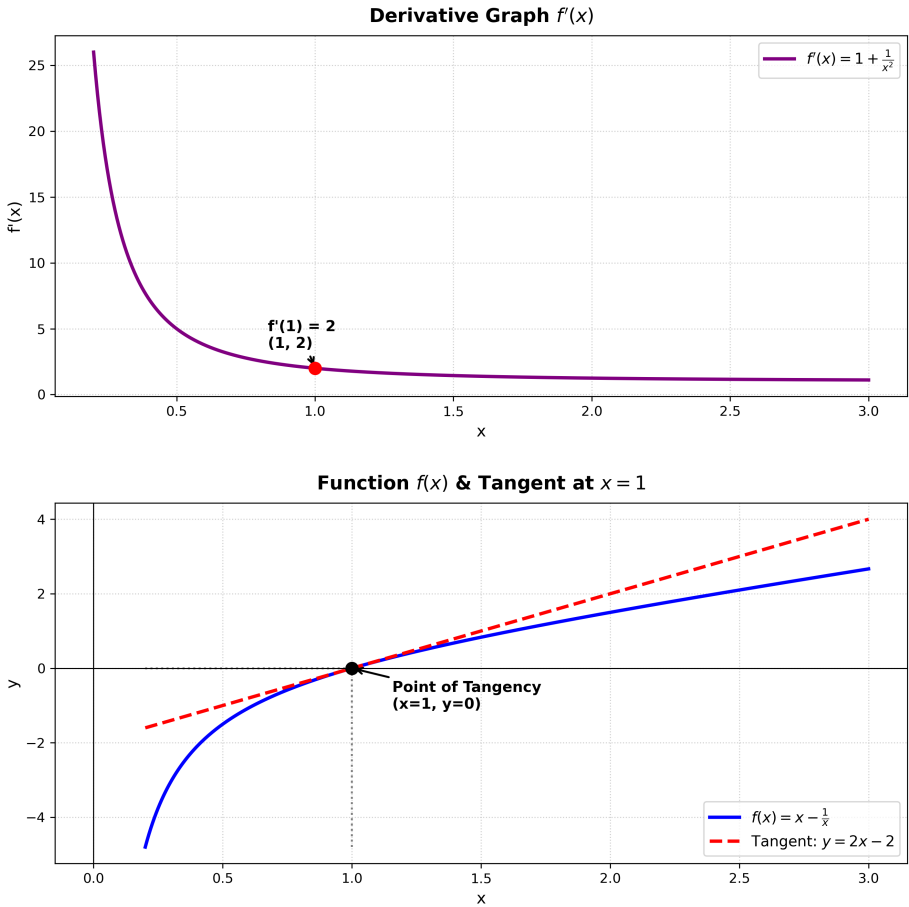


Fig. 1.0.8.1. Derivative vs x ; function vs x with tangent

codes/bm8.py

1.0.9 The elements of the dataset $\{-5, 1, a, 5, b\}$ are in ascending order. If both mean and median of the dataset are equal to 3, what is the value of b ? (GATE BM 2025)

- (a) 7
- (b) 11
- (c) 9
- (d) 12

Solution:

The dataset consists of 5 elements arranged in ascending order:

$$-5 \leq 1 \leq a \leq 5 \leq b \quad (1.0.9.1)$$

Since the dataset has an odd number of elements ($n = 5$), the median is the middle (3rd) element:

$$\text{Median} = a = 3 \quad (1.0.9.2)$$

The mean of the dataset is given as 3:

$$\text{Mean} = \frac{-5 + 1 + a + 5 + b}{5} = 3 \quad (1.0.9.3)$$

Substituting $a = 3$ into equation (9.3):

$$\frac{-5 + 1 + 3 + 5 + b}{5} = 3 \quad (1.0.9.4)$$

$$\frac{4 + b}{5} = 3 \quad (1.0.9.5)$$

$$4 + b = 15 \quad (1.0.9.6)$$

$$b = 11 \quad (1.0.9.7)$$

$$\text{Hence, the correct answer is (b) 11} \quad (1.0.9.8)$$

codes/bm9.py

1.0.10 A time domain signal is given by

$$x(t) = 20 \sin(100\pi t) + 36 \sin(150\pi t) - 2 \sin(300\pi t), \quad (1.0.10.1)$$

where time t is in seconds. Among the following options, what is the frequency (in Hz) at which $x(t)$ should be sampled for accurate signal reconstruction? (GATE BM 2025)

- (a) 500
- (b) 150
- (c) 200
- (d) 250

Solution:

To avoid aliasing and accurately reconstruct the signal, the sampling frequency f_s

must satisfy the Nyquist sampling criterion:

$$f_s \geq 2f_{\max} \quad (1.0.10.2)$$

where $f_{\max}$ is the highest frequency component in the signal $x(t)$.

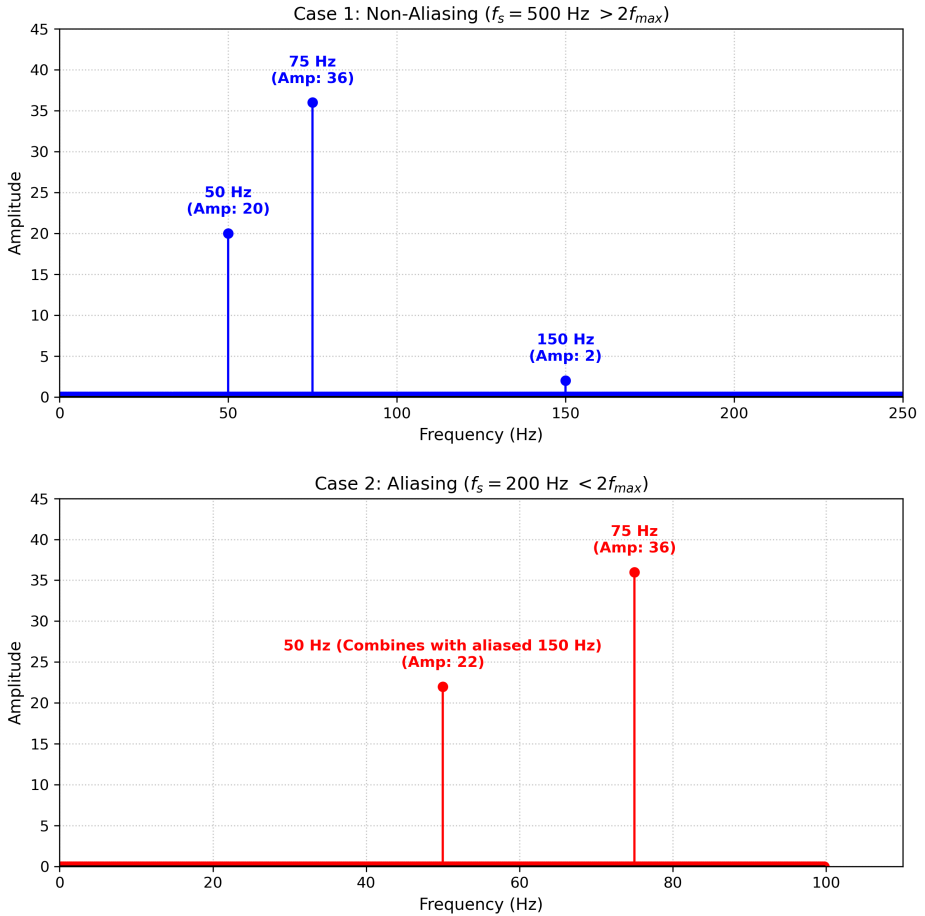


Fig. 1.0.10.1. Aliasing and Non-Aliasing

codes/bm10a.py

The signal consists of three sinusoidal components with angular frequencies ω_1 , ω_2 , and ω_3 :

$$\omega_1 = 100\pi \Rightarrow f_1 = \frac{100\pi}{2\pi} = 50 \text{ Hz} \quad (1.0.10.3)$$

$$\omega_2 = 150\pi \Rightarrow f_2 = \frac{150\pi}{2\pi} = 75 \text{ Hz} \quad (1.0.10.4)$$

$$\omega_3 = 300\pi \Rightarrow f_3 = \frac{300\pi}{2\pi} = 150 \text{ Hz} \quad (1.0.10.5)$$

The maximum frequency component present in the signal is:

$$f_{\max} = 150 \text{ Hz} \quad (1.0.10.6)$$

According to the Nyquist theorem, the minimum required sampling rate (Nyquist rate) is:

$$f_s \geq 2 \times 150 \text{ Hz} = 300 \text{ Hz} \quad (1.0.10.7)$$

Among the given options, the only frequency greater than or equal to 300 Hz that guarantees accurate signal reconstruction is 500 Hz.

Hence, the correct answer is (a) 500

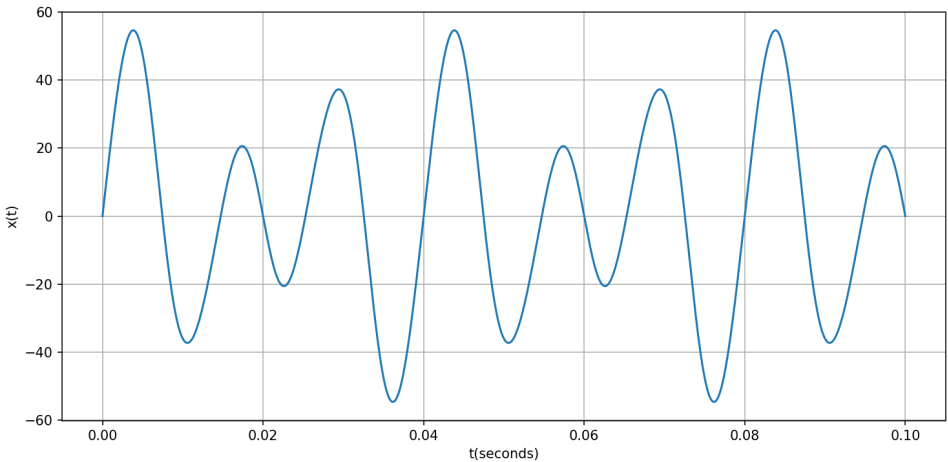


Fig. 1.0.10.2. Plot of $x(t)$ vs time

codes/bm10.py

- 1.0.11 A voltage source $V_s = 5 \sin(100t + \frac{\pi}{2})$ volts is driving a current $I = 10 \sin(100t + \frac{\pi}{6})$ mA through a circuit. The average power dissipated in the circuit is _____ mW. (GATE BM 2025)

- (a) 0.0
- (b) 12.5
- (c) 25.0
- (d) 50.0

Solution:

The average power P_{avg} dissipated in an AC circuit in terms of peak voltage V_0 and peak current I_0 is given by:

$$P_{\text{avg}} = \frac{1}{2} V_0 I_0 \cos(\phi_v - \phi_i) \quad (1.0.11.1)$$

Given data from the problem:

$$V_s(t) = 5 \sin \left(100t + \frac{\pi}{2} \right) \text{ V}$$

$$I(t) = 10 \sin \left(100t + \frac{\pi}{6} \right) \text{ mA}$$

Extracting the peak values and phase angles:

$$V_0 = 5 \text{ V}$$

$$I_0 = 10 \text{ mA}$$

$$\phi_v = \frac{\pi}{2} \text{ rad}$$

$$\phi_i = \frac{\pi}{6} \text{ rad}$$

Calculate the phase difference θ :

$$\theta = \phi_v - \phi_i = \frac{\pi}{2} - \frac{\pi}{6} = \frac{3\pi - \pi}{6} = \frac{2\pi}{6} = \frac{\pi}{3} \text{ rad} \quad (1.0.11.2)$$

Substituting these values into the average power formula:

$$P_{\text{avg}} = \frac{1}{2} \times 5 \times 10 \times \cos \left(\frac{\pi}{3} \right) \text{ mW}$$

$$P_{\text{avg}} = \frac{1}{2} \times 50 \times \frac{1}{2} \text{ mW}$$

$$P_{\text{avg}} = \frac{50}{4} \text{ mW}$$

$$P_{\text{avg}} = 12.5 \text{ mW}$$

Correct Option: **(b) 12.5**

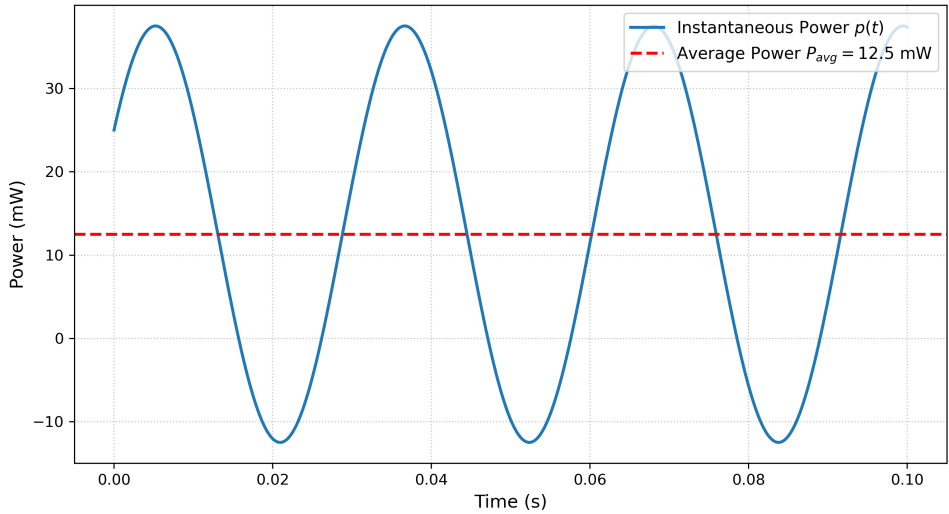


Fig. 1.0.11.1. Power vs time

codes/bm11.py

1.0.12 N independent trials of a binomial random variable yield an expectation value of 16 and variance of 12. What is the value of N ? (GATE BM 2025)

- (a) 80
- (b) 60
- (c) 45
- (d) 64

Solution:

For a Binomial distribution $X \sim \mathcal{B}(N, p)$ with N independent trials and success probability p :

The expectation value (mean μ) and variance (σ^2) are given by:

$$\mu = E(X) = Np = 16 \quad (1.0.12.1)$$

$$\sigma^2 = \text{Var}(X) = Np(1 - p) = 12 \quad (1.0.12.2)$$

To approximate this Binomial distribution using a Gaussian (Normal) distribution $Y \sim \mathcal{N}(\mu, \sigma^2)$ with the same mean and variance, the Probability Density Function (PDF) is defined as:

$$f_Y(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \quad (1.0.12.3)$$

Substituting the given mean $\mu = 16$ and variance $\sigma^2 = 12$:

$$f_Y(x) = \frac{1}{\sqrt{24\pi}} \exp\left(-\frac{(x-16)^2}{24}\right) \quad (1.0.12.4)$$

To find N , divide the variance equation by the mean equation:

$$\frac{Np(1-p)}{Np} = \frac{12}{16} \quad (1.0.12.5)$$

Simplifying:

$$1-p = \frac{3}{4} = 0.75 \quad (1.0.12.6)$$

Solving for p :

$$p = 1 - 0.75 = 0.25 = \frac{1}{4} \quad (1.0.12.7)$$

Substituting $p = \frac{1}{4}$ back into the expectation value equation:

$$N\left(\frac{1}{4}\right) = 16 \quad (1.0.12.8)$$

$$N = 16 \times 4 \quad (1.0.12.9)$$

$$N = 64 \quad (1.0.12.10)$$

Correct Option: (d) 64

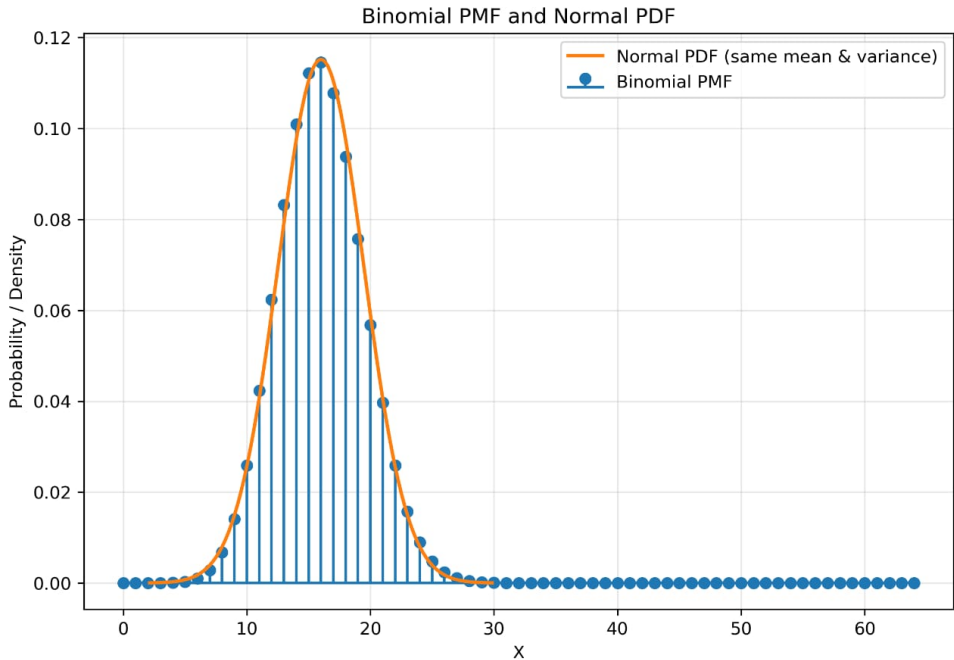


Fig. 1.0.12.1. Probability vs values

codes/bm12.py

1.0.13 Given a function $y(x)$ satisfying the differential equation

$$y'' - 0.25y = 0, \quad (1.0.13.1)$$

with initial conditions $y(0) = 1$; $y'(0) = 1$, what is the value of $y(\log_e 100)$?

Here, y' and y'' are the first and second derivatives of y , respectively.
(GATE BM 2025)

- (a) 14.05
- (b) 14.25
- (c) 14.95
- (d) 14.65

Solution:

Given the second-order linear differential equation:

$$y''(x) - 0.25y(x) = 0 \quad (1.0.13.2)$$

with initial conditions $y(0) = 1$ and $y'(0) = 1$.

Take the Laplace Transform on both sides

Let $\mathcal{L}\{y(x)\} = Y(s)$. Applying the Laplace Transform derivative property:

$$\mathcal{L}\{y''(x)\} = s^2Y(s) - sy(0) - y'(0)$$

Substituting into the differential equation gives:

$$[s^2Y(s) - sy(0) - y'(0)] - 0.25Y(s) = 0$$

Substitute the initial conditions

Substitute $y(0) = 1$ and $y'(0) = 1$:

$$s^2Y(s) - s - 1 - 0.25Y(s) = 0$$

Group $Y(s)$ terms together:

$$(s^2 - 0.25)Y(s) = s + 1$$

$$Y(s) = \frac{s + 1}{s^2 - 0.25} = \frac{s + 1}{\left(s - \frac{1}{2}\right)\left(s + \frac{1}{2}\right)}$$

Apply Partial Fraction Decomposition

Express $Y(s)$ as:

$$Y(s) = \frac{A}{s - 0.5} + \frac{B}{s + 0.5}$$

Solving for constants A and B :

$$A = \lim_{s \rightarrow 0.5} (s - 0.5)Y(s) = \frac{0.5 + 1}{0.5 + 0.5} = 1.5 = \frac{3}{2}$$

$$B = \lim_{s \rightarrow -0.5} (s + 0.5)Y(s) = \frac{-0.5 + 1}{-0.5 - 0.5} = -0.5 = -\frac{1}{2}$$

So, the transform is:

$$Y(s) = \frac{1.5}{s - 0.5} - \frac{0.5}{s + 0.5}$$

Take the Inverse Laplace Transform

Using $\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$:

$$y(x) = 1.5e^{0.5x} - 0.5e^{-0.5x}$$

Evaluate $y(\log_e 100)$

Substitute $x = \log_e 100 = \ln(100)$:

$$e^{0.5 \ln(100)} = e^{\ln(100^{0.5})} = \sqrt{100} = 10$$

$$e^{-0.5 \ln(100)} = \frac{1}{\sqrt{100}} = \frac{1}{10} = 0.1$$

Now, evaluate $y(\ln 100)$:

$$y(\ln 100) = 1.5(10) - 0.5(0.1) = 15 - 0.05 = 14.95$$

Correct Option: (c) 14.95

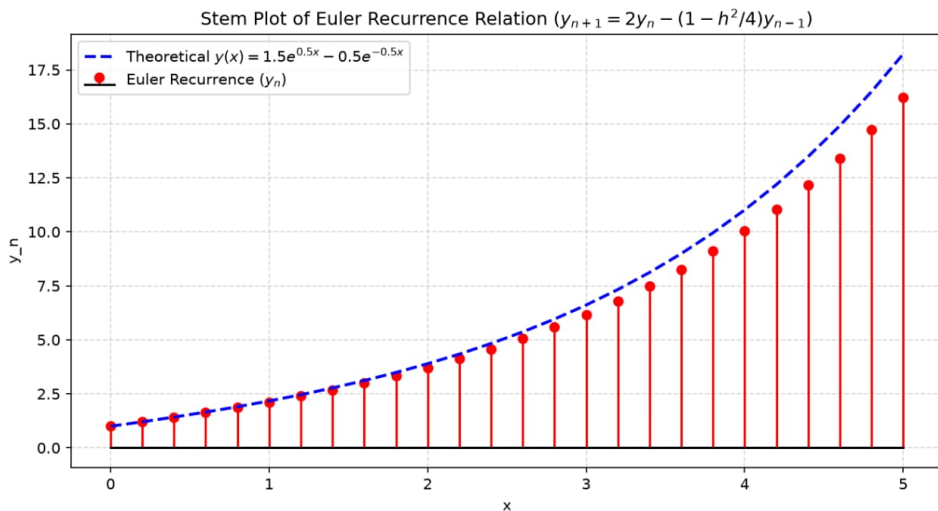
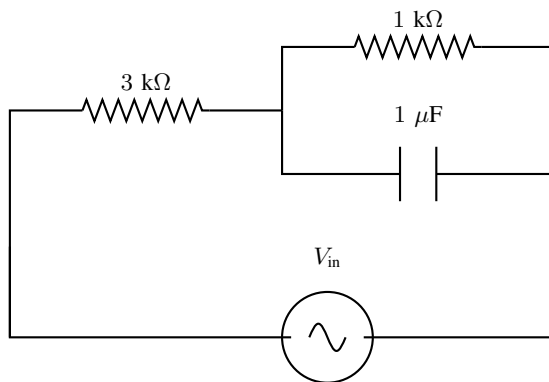


Fig. 1.0.13.1. Stem plot of Euler Recurrence relation

codes/bm13.py

- 1.0.14 In the circuit shown below, $V_{in} = 10 \cos(1000t)$ volts. The magnitude of the input impedance of the circuit is _____ $k\Omega$. (rounded off to one decimal place)



(GATE BM 2025)

- (a) 5.0
- (b) 3.5
- (c) 6.0
- (d) 4.5

Solution:

Given the input voltage $V_{in} = 10 \cos(1000t)$ V, the circuit parameter values are:

$$R_1 = 3 \text{ k}\Omega, \quad R_2 = 1 \text{ k}\Omega, \quad C = 1 \mu\text{F}, \quad \omega = 1000 \text{ rad/s}$$

Symbolic Impedance Derivation

The impedance of the capacitor in terms of angular frequency ω is:

$$Z_C(\omega) = \frac{1}{j\omega C} = -\frac{j}{\omega C}$$

The parallel combination of resistor R_2 and capacitor $Z_C(\omega)$ gives $Z_p(\omega)$:

$$Z_p(\omega) = \frac{R_2 \cdot Z_C(\omega)}{R_2 + Z_C(\omega)} = \frac{R_2 \left(-\frac{j}{\omega C}\right)}{R_2 - \frac{j}{\omega C}} = \frac{-jR_2}{\omega R_2 C - j}$$

Multiplying numerator and denominator by the complex conjugate $(\omega R_2 C + j)$:

$$Z_p(\omega) = \frac{-jR_2(\omega R_2 C + j)}{(\omega R_2 C - j)(\omega R_2 C + j)} = \frac{R_2 - j\omega R_2^2 C}{1 + (\omega R_2 C)^2}$$

The total input impedance $Z_{in}(\omega)$ in series with R_1 is:

$$Z_{in}(\omega) = R_1 + Z_p(\omega) = \left(R_1 + \frac{R_2}{1 + (\omega R_2 C)^2} \right) - j \left(\frac{\omega R_2^2 C}{1 + (\omega R_2 C)^2} \right)$$

Magnitude and Phase Expressions in terms of ω

1. Magnitude Spectrum $|Z_{in}(\omega)|$:

$$|Z_{in}(\omega)| = \sqrt{\left[R_1 + \frac{R_2}{1 + (\omega R_2 C)^2} \right]^2 + \left[\frac{\omega R_2^2 C}{1 + (\omega R_2 C)^2} \right]^2}$$

Simplifying the algebraic expression:

$$|Z_{in}(\omega)| = \sqrt{R_1^2 + \frac{2R_1 R_2 + R_2^2 + (\omega R_1 R_2 C)^2}{1 + (\omega R_2 C)^2}}$$

2. Phase Spectrum $\angle Z_{in}(\omega)$:

$$\angle Z_{in}(\omega) = -\tan^{-1} \left(\frac{\frac{\omega R_2^2 C}{1 + (\omega R_2 C)^2}}{R_1 + \frac{R_2}{1 + (\omega R_2 C)^2}} \right) = -\tan^{-1} \left(\frac{\omega R_2^2 C}{R_1(1 + (\omega R_2 C)^2) + R_2} \right)$$

Numerical Evaluation at $\omega = 1000$ rad/s

Substituting $R_1 = 3 \text{ k}\Omega$, $R_2 = 1 \text{ k}\Omega$, $C = 1 \mu\text{F}$, and $\omega = 1000$ rad/s:

First calculate the normalized frequency factor:

$$\omega R_2 C = (1000)(1 \times 10^3)(1 \times 10^{-6}) = 1$$

Now, compute real and imaginary components of Z_p :

$$Z_p = \frac{1 - j(1)}{1 + (1)^2} = 0.5 - j0.5 \text{ k}\Omega$$

Total impedance:

$$Z_{\text{in}} = 3 + (0.5 - j0.5) = 3.5 - j0.5 \text{ k}\Omega$$

Evaluating Magnitude $|Z_{\text{in}}|$:

$$|Z_{\text{in}}| = \sqrt{(3.5)^2 + (-0.5)^2} = \sqrt{12.25 + 0.25} = \sqrt{12.5} \approx 3.5355 \text{ k}\Omega \approx 3.5 \text{ k}\Omega$$

Evaluating Phase $\angle Z_{\text{in}}$:

$$\angle Z_{\text{in}} = \tan^{-1} \left(\frac{-0.5}{3.5} \right) = -\tan^{-1} \left(\frac{1}{7} \right) \approx -8.13^\circ$$

Correct Answer: (b) 3.5

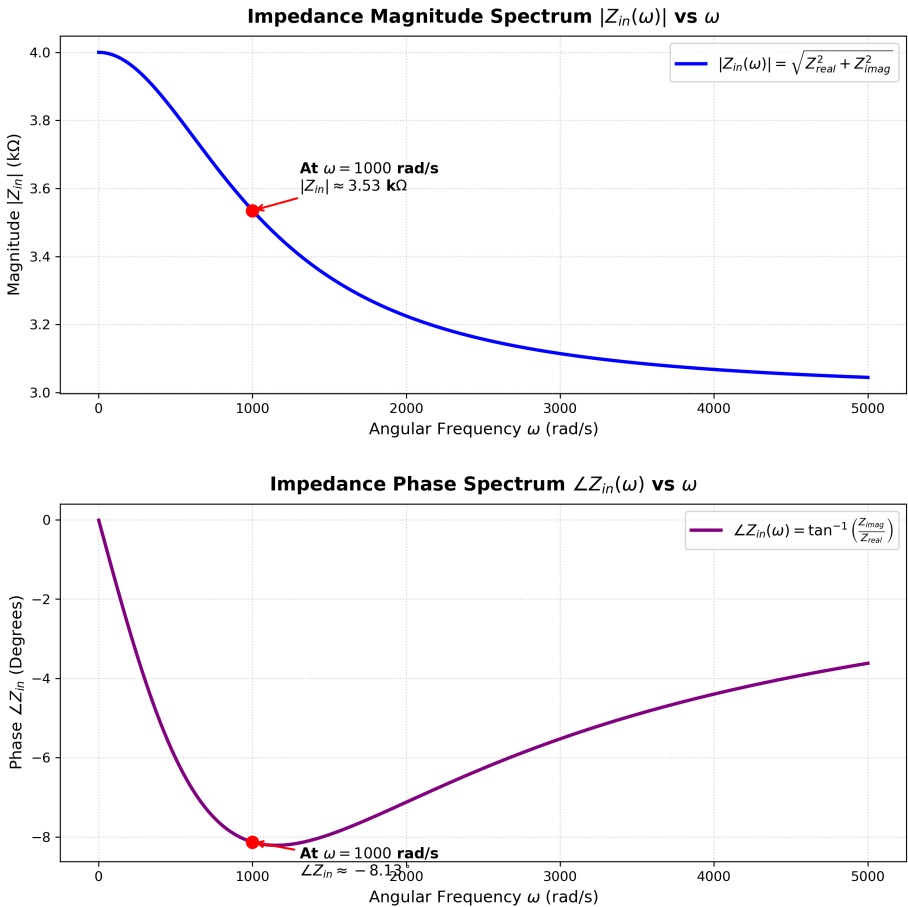


Fig. 1.0.14.1. Impedance Magnitude VS Omega; Phase VS Omega

codes/bm14.py

1.0.15 Which of the following is/are CORRECT about the value of $y = \log_e(-e^x)$, where x is a real number. (GATE BM 2025)

- (a) y is real valued
- (b) y is complex valued
- (c) $y = x \pm in\pi$, where n is an odd integer and $i = \sqrt{-1}$
- (d) $y = x \pm in\pi$, where n is an even integer and $i = \sqrt{-1}$

Solution:

Given the expression:

$$y = \log_e(-e^x) = \ln(-e^x) \quad (1.0.15.1)$$

Using logarithmic properties, we rewrite the term inside the logarithm:

$$-e^x = (-1) \cdot e^x \quad (1.0.15.2)$$

Applying Euler's identity, -1 can be expressed in complex exponential form as:

$$-1 = e^{i(2k+1)\pi}, \quad \text{for } k \in \mathbb{Z} \quad (1.0.15.3)$$

Substituting equation (15.3) into equation (15.2):

$$-e^x = e^x \cdot e^{i(2k+1)\pi} = e^{x+i(2k+1)\pi} \quad (1.0.15.4)$$

Taking the natural logarithm on both sides:

$$y = \ln \left(e^{x+i(2k+1)\pi} \right) \quad (1.0.15.5)$$

$$y = x + i(2k+1)\pi, \quad \text{for } k \in \mathbb{Z} \quad (1.0.15.6)$$

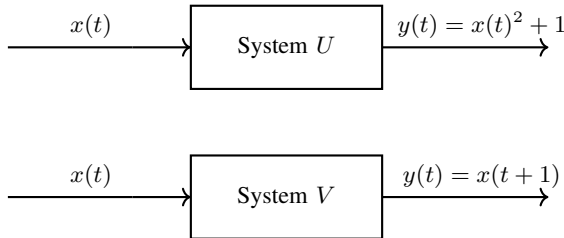
Let $n = |2k+1|$. Since $2k+1$ is always an odd integer, n represents an odd integer:

$$y = x \pm in\pi, \quad \text{where } n \text{ is an odd integer and } i = \sqrt{-1} \quad (1.0.15.7)$$

Since y contains an imaginary component $in\pi$ (with $n \geq 1$), y is a complex-valued function.

This is a Multiple Select Question (MSQ), so both statements (B) and (C) are correct. Hence, the correct options are (b), (c)

1.0.16 Two systems U and V are defined as shown below. Which of the following statement(s) is/are CORRECT?



(GATE BM 2025)

- (a) System U is nonlinear; System V is linear
- (b) System U is causal; System V is noncausal

- (c) System U is noncausal; System V is causal
 (d) System U is nonlinear; System V is nonlinear

$$\text{System } U : y(t) = x(t)^2 + 1$$

$$\text{System } V : y(t) = x(t + 1)$$

$$\text{System } U: y(t) = x(t)^2 + 1$$

1. Linearity Check (Additive & Homogeneous):

- Additivity: Let $y_1(t) = x_1(t)^2 + 1$ and $y_2(t) = x_2(t)^2 + 1$. For input $x_3(t) = x_1(t) + x_2(t)$, the output is:

$$y_3(t) = [x_1(t) + x_2(t)]^2 + 1 = x_1(t)^2 + 2x_1(t)x_2(t) + x_2(t)^2 + 1$$

Since $y_1(t) + y_2(t) = x_1(t)^2 + x_2(t)^2 + 2 \neq y_3(t)$, the system is not additive.

- Homogeneity: For input $a \cdot x(t)$, the output is:

$$y_{scaled}(t) = [a \cdot x(t)]^2 + 1 = a^2 x(t)^2 + 1 \neq a \cdot y(t)$$

Thus, it is not homogeneous.

Conclusion: System U is nonlinear.

2. Causality Check:

- At $t = 0$: $y(0) = x(0)^2 + 1$.
- The output at $t = 0$ depends only on the present input $x(0)$.

Conclusion: System U is causal.

$$\text{System } V: y(t) = x(t + 1)$$

1. Linearity Check (Additive & Homogeneous):

- Additivity: Let $y_1(t) = x_1(t + 1)$ and $y_2(t) = x_2(t + 1)$. For input $x_3(t) = x_1(t) + x_2(t)$:

$$y_3(t) = x_1(t + 1) + x_2(t + 1) = y_1(t) + y_2(t) \quad (\text{Satisfied})$$

- Homogeneity: For input $a \cdot x(t)$:

$$y_{scaled}(t) = a \cdot x(t + 1) = a \cdot y(t) \quad (\text{Satisfied})$$

Conclusion: System V is linear.

2. Causality Check:

- At $t = 0$: $y(0) = x(0 + 1) = x(1)$.
- The output at $t = 0$ depends on the future input $x(1)$.

Conclusion: System V is noncausal.

Option (a): System U is nonlinear; System V is linear

- System U fails additivity and homogeneity $\rightarrow$ nonlinear.

- System V satisfies additivity and homogeneity $\rightarrow$ linear.

Option (b): System U is causal; System V is noncausal

- For System U , $y(0) = x(0)^2 + 1$ (depends on current input) $\rightarrow$ causal.
- For System V , $y(0) = x(1)$ (depends on future input) $\rightarrow$ noncausal.

Option (c): System U is noncausal; System V is causal

- Incorrect based on the $y(0)$ check above.

Option (d): System U is nonlinear; System V is nonlinear

- System V is linear, making this statement incorrect.

Correct options are (a) and (b).

1.0.17 A discrete linear time invariant system has an impulse response function given by

$$h[n] = \delta[n] + \frac{1}{2}\delta[n-1] + \frac{1}{3}\delta[n-2]. \quad (1.0.17.1)$$

For input signal $x[n] = \delta[n] + \delta[n-1]$, which of the following option(s) is/are CORRECT for the output signal $y[n]$? (GATE BM 2025)

- (a) $y[2] = 11/6$
- (b) $y[2] = 5/6$
- (c) $y[k] = 0$ for all $k \geq 3$
- (d) $y[k] = 0$ for all $k \geq 4$

Solution:

Express sequences in vector form

The impulse response sequence $h[n]$ of length $M = 3$ is:

$$h[n] = \left[1, \frac{1}{2}, \frac{1}{3}\right] \quad \text{for } n = 0, 1, 2$$

The input signal sequence $x[n]$ of length $N = 2$ is:

$$x[n] = [1, 1] \quad \text{for } n = 0, 1$$

The linear convolution $y[n] = h[n] * x[n]$ has a total length of $L = M + N - 1 = 3 + 2 - 1 = 4$ points (spanning $n = 0, 1, 2, 3$).

Construct the Convolution (Toeplitz) Matrix

Linear convolution can be represented as a matrix-vector product $\mathbf{y} = \mathbf{H}\mathbf{x}$, where $\mathbf{H}$ is a $(M + N - 1) \times N = 4 \times 2$ lower-triangular Toeplitz matrix formed by shifting $h[n]$ downwards:

$$\mathbf{H} = \begin{bmatrix} h[0] & 0 \\ h[1] & h[0] \\ h[2] & h[1] \\ 0 & h[2] \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \\ \frac{1}{3} & \frac{1}{2} \\ 0 & \frac{1}{3} \end{bmatrix}$$

The input vector $\mathbf{x}$ is:

$$\mathbf{x} = \begin{bmatrix} x[0] \\ x[1] \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Matrix-Vector Multiplication

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \\ \frac{1}{3} & \frac{1}{2} \\ 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Computing each element:

$$y[0] = 1(1) + 0(1) = 1$$

$$y[1] = \frac{1}{2}(1) + 1(1) = \frac{3}{2}$$

$$y[2] = \frac{1}{3}(1) + \frac{1}{2}(1) = \frac{5}{6}$$

$$y[3] = 0(1) + \frac{1}{3}(1) = \frac{1}{3}$$

For $k \geq 4$, $y[k] = 0$ since the signal duration ends at $n = 3$.

Verify the Options

- Option (a): $y[2] = \frac{11}{6}$ — Incorrect
- Option (b): $y[2] = \frac{5}{6}$ — Correct
- Option (c): $y[k] = 0$ for all $k \geq 3$ — Incorrect (since $y[3] = \frac{1}{3} \neq 0$)
- Option (d): $y[k] = 0$ for all $k \geq 4$ — Correct

Correct Options: (b) and (d)

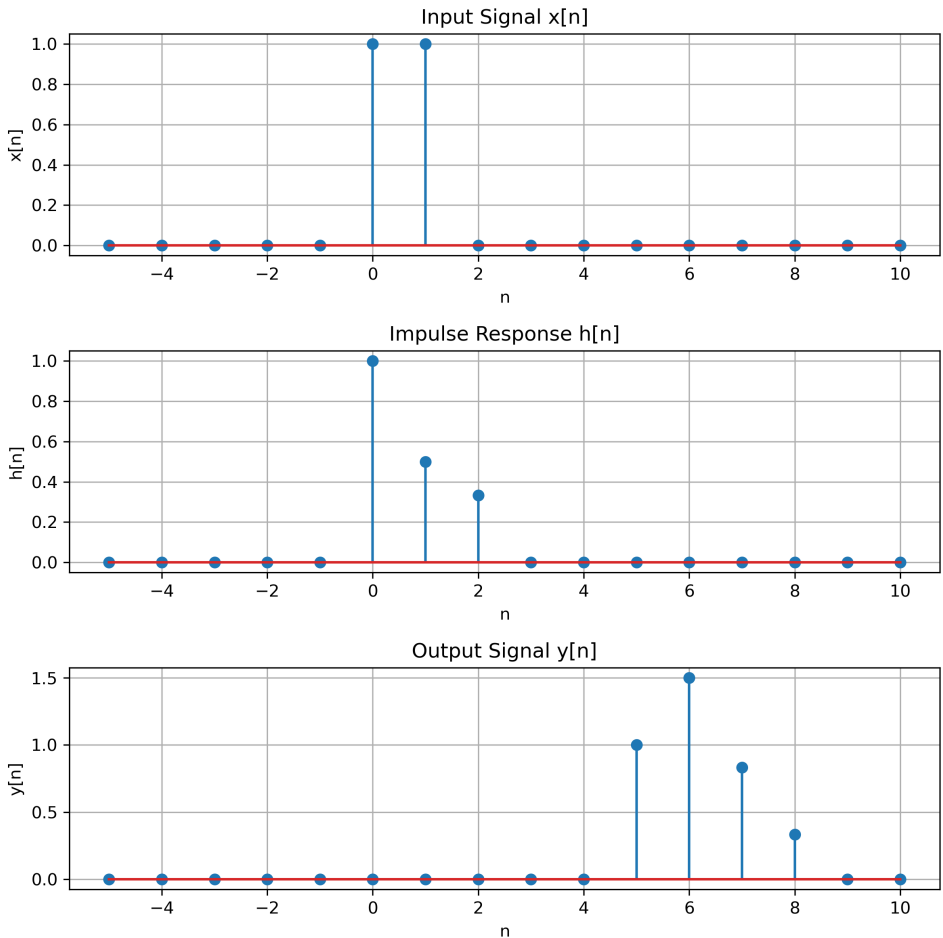


Fig. 1.0.17.1. $x[n], h[n], y[n]$ vs n

codes/bm17a.py

- 1.0.18 For an 8-bit Digital to Analog Converter (DAC), the binary input 0000 0000 results in 0 V, and 1111 1111 results in 5 V. The output of the DAC for an input of 1011 0111 is _____ (in volts, rounded off to two decimal places). (GATE BM 2025)

Solution:

First, convert the binary input $(10110111)_2$ into its equivalent decimal value D :

$$D = 1 \cdot 2^7 + 0 \cdot 2^6 + 1 \cdot 2^5 + 1 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 \quad (1.0.18.1)$$

$$D = 128 + 0 + 32 + 16 + 0 + 4 + 2 + 1 = 183 \quad (1.0.18.2)$$

For an 8-bit DAC:

- Minimum digital value $(00000000)_2 = 0$ corresponds to 0 V.
- Maximum digital value $(11111111)_2 = 255$ corresponds to Full-Scale Voltage $V_{FS} = 5$ V.

The output voltage V_{out} for a given decimal value D is given by:

$$V_{out} = V_{FS} \times \left(\frac{D}{2^n - 1} \right) \quad (1.0.18.3)$$

where $n = 8$ bits, so $2^8 - 1 = 255$.

Substituting $D = 183$ and $V_{FS} = 5$ V:

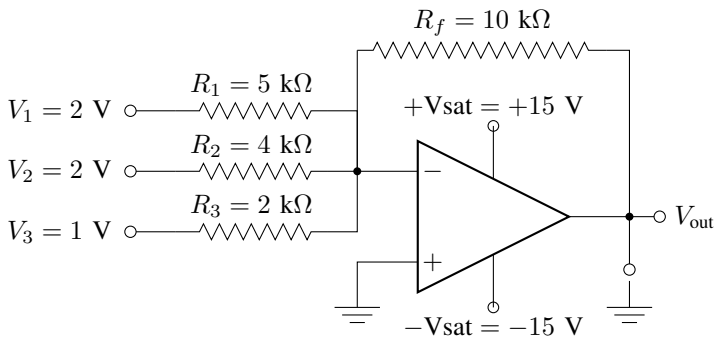
$$V_{out} = 5 \times \left(\frac{183}{255} \right) \quad (1.0.18.4)$$

$$V_{out} = \frac{915}{255} \approx 3.5882 \text{ V} \quad (1.0.18.5)$$

Rounding off to two decimal places gives: 3.59

codes/bm18.py

1.0.19 What is the output voltage V_{out} for the circuit shown below? _____ (in volts, rounded off to the nearest integer).



(GATE BM 2025)

Solution:

The given operational amplifier circuit is an **inverting summing amplifier**.

The general expression for the output voltage of an inverting summing amplifier is given by:

$$V_{\text{out}} = - \left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right) \quad (1.0.19.1)$$

Given values:

- $R_f = 10 \text{ k}\Omega$
- $R_1 = 5 \text{ k}\Omega, \quad V_1 = 2 \text{ V}$
- $R_2 = 4 \text{ k}\Omega, \quad V_2 = 2 \text{ V}$
- $R_3 = 2 \text{ k}\Omega, \quad V_3 = 1 \text{ V}$
- Power supply saturation limits: $\pm V_{\text{sat}} = \pm 15 \text{ V}$

Substitute the given component values into equation (19.1):

$$V_{\text{out}} = - \left(\frac{10}{5}(2) + \frac{10}{4}(2) + \frac{10}{2}(1) \right) \quad (1.0.19.2)$$

$$V_{\text{out}} = - (2(2) + 2.5(2) + 5(1)) \quad (1.0.19.3)$$

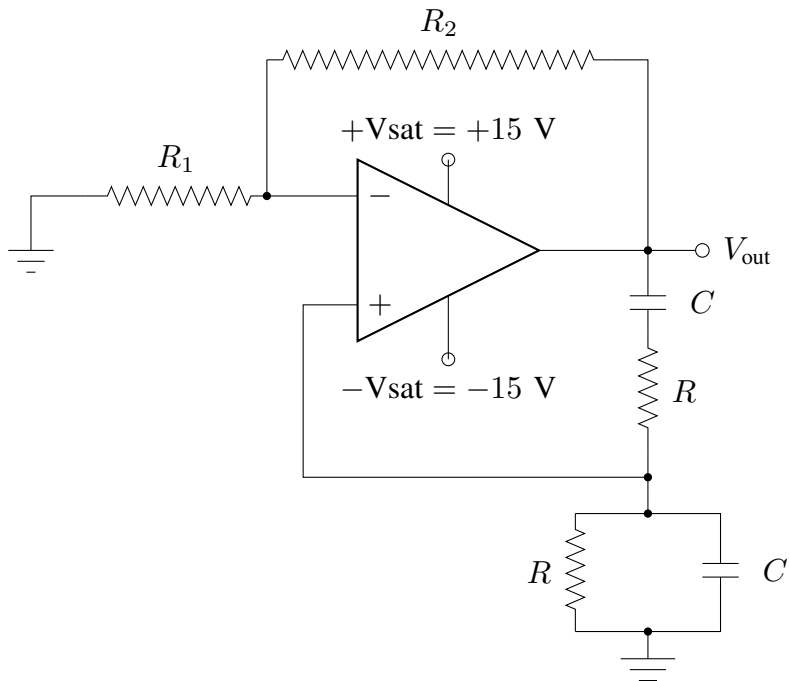
$$V_{\text{out}} = -(4 + 5 + 5) = -14 \text{ V} \quad (1.0.19.4)$$

Since the calculated voltage (-14 V) lies within the saturation limits (-15 V to $+15 \text{ V}$), the op-amp stays in its linear region.

Rounding off to the nearest integer gives: -14

1.0.20 The frequency of the oscillator circuit shown in the figure below is _____ kHz. (rounded off to two decimal places)

Given: $R = 1 \text{ k}\Omega$; $R_1 = 2 \text{ k}\Omega$; $R_2 = 6 \text{ k}\Omega$; $C = 0.1 \text{ }\mu\text{F}$



(GATE BM 2025)

Solution:

The given circuit is a ****Wien Bridge Oscillator****.

The frequency of oscillation f_0 for a Wien Bridge Oscillator constructed using identical frequency-selective resistors R and capacitors C is given by:

$$f_0 = \frac{1}{2\pi RC} \quad (1.0.20.1)$$

Given values:

- $R = 1 \text{ k}\Omega = 10^3 \Omega$
- $C = 0.1 \mu\text{F} = 0.1 \times 10^{-6} \text{ F} = 10^{-7} \text{ F}$

Substitute R and C into equation (20.1):

$$f_0 = \frac{1}{2\pi(10^3)(10^{-7})} = \frac{1}{2\pi \times 10^{-4}} \text{ Hz} \quad (1.0.20.2)$$

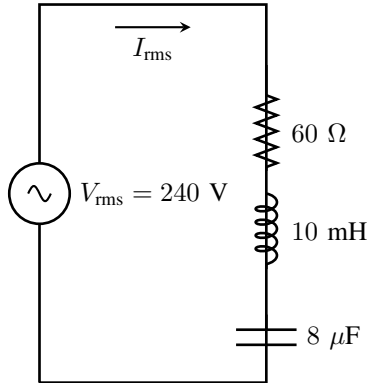
$$f_0 = \frac{10^4}{2\pi} \text{ Hz} = \frac{10000}{2\pi} \approx 1591.549 \text{ Hz} \quad (1.0.20.3)$$

Converting to kilohertz (kHz):

$$f_0 = \frac{1591.549}{1000} \text{ kHz} \approx 1.5915 \text{ kHz} \quad (1.0.20.4)$$

Rounding off to two decimal places: 1.59

- 1.0.21 For the RLC circuit shown below, the root mean square current (I_{rms}) at the resonance frequency is _____ amperes. (rounded off to the nearest integer)



(GATE BM 2025)

Solution:

Time-Domain Differential Equation

For the series RLC circuit driven by a sinusoidal voltage source $v(t) = V_m \cos(\omega t)$, Kirchhoff's Voltage Law (KVL) gives:

$$Ri(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int i(t) dt = v(t)$$

Assuming steady-state sinusoidal response $i(t) = I_m \cos(\omega t - \phi)$:

Phasor / Frequency-Domain Impedance

Converting to complex phasor notation $V(t) = V_{\text{rms}} \sqrt{2} e^{j\omega t}$ and $I(t) = I_{\text{rms}} \sqrt{2} e^{j(\omega t - \phi)}$:

$$Z(\omega) = \frac{V(\omega)}{I(\omega)} = R + j\omega L + \frac{1}{j\omega C} = R + j \left(\omega L - \frac{1}{\omega C} \right)$$

The magnitude of the circuit impedance as a function of angular frequency ω is:

$$|Z(\omega)| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}$$

Resonance Condition

Resonance occurs at frequency ω_0 where the reactive components cancel each other

($X_L = X_C$), minimizing $|Z(\omega)|$:

$$\omega_0 L - \frac{1}{\omega_0 C} = 0 \implies \omega_0 = \frac{1}{\sqrt{LC}}$$

Evaluating the resonance angular frequency and cyclic frequency:

$$\omega_0 = \frac{1}{\sqrt{(10 \times 10^{-3} \text{ H})(8 \times 10^{-6} \text{ F})}} = \frac{1}{\sqrt{8 \times 10^{-8}}} = 3535.53 \text{ rad/s}$$

$$f_0 = \frac{\omega_0}{2\pi} \approx 562.7 \text{ Hz}$$

At $\omega = \omega_0$, the net reactance is zero ($Z(\omega_0) = R = 60 \Omega$).

RMS Current Evaluation at Resonance

The root-mean-square (RMS) current I_{rms} at resonance is:

$$I_{\text{rms}}(\omega_0) = \frac{V_{\text{rms}}}{|Z(\omega_0)|} = \frac{V_{\text{rms}}}{R}$$

Substituting the given numerical values ($V_{\text{rms}} = 240 \text{ V}$, $R = 60 \Omega$):

$$I_{\text{rms}}(\omega_0) = \frac{240 \text{ V}}{60 \Omega} = 4 \text{ A}$$

Answer: 4

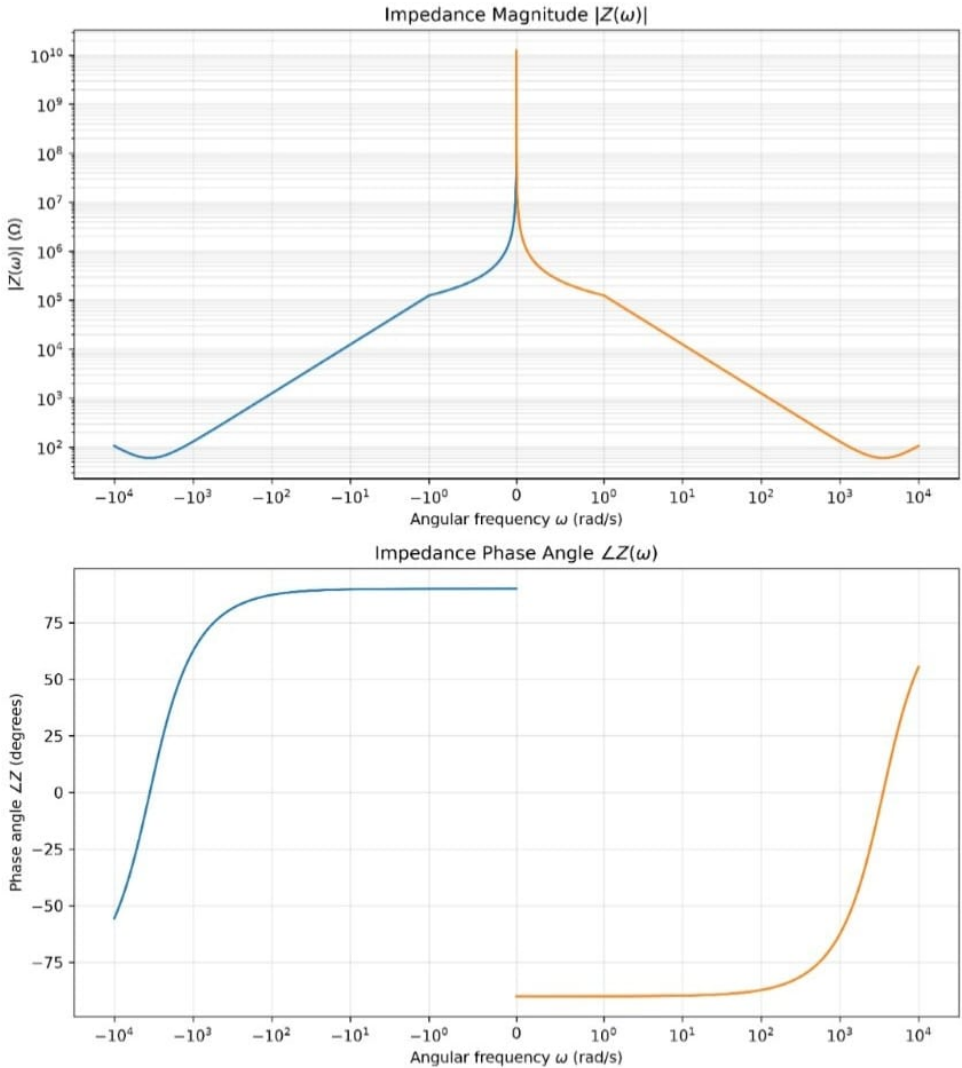


Fig. 1.0.21.1. Impedance mag vs angular frequency, Impedance phase angle vs angular frequency

codes/bm21.py

- 1.0.22 Newton Raphson method is used to solve the equation $x^3 - 2x - 5 = 0$ with an initial guess $x_0 = 3$. What is x_1 , the value after ONE iteration? _____ (rounded off to two decimal places) (GATE BM 2025)

Solution:

Function Definition & Derivative

Given the non-linear equation:

$$f(x) = x^3 - 2x - 5 = 0$$

Taking the derivative with respect to x :

$$f'(x) = 3x^2 - 2$$

General Newton-Raphson Update Equation

The general iterative update formula is defined as:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Substituting $f(x_n)$ and $f'(x_n)$ into the update equation gives the explicit recurrence relation:

$$x_{n+1} = x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2} = \frac{2x_n^3 + 5}{3x_n^2 - 2}$$

First Iteration Calculation ($n = 0$ with $x_0 = 3$)

Evaluating $f(x_0)$ and $f'(x_0)$:

$$f(3) = (3)^3 - 2(3) - 5 = 27 - 6 - 5 = 16$$

$$f'(3) = 3(3)^2 - 2 = 27 - 2 = 25$$

Using the update equation to calculate x_1 :

$$x_1 = 3 - \frac{16}{25} = 3 - 0.64 = 2.36$$

Subsequent Iterations (Finding the Actual Root)

Using the simplified update equation $x_{n+1} = \frac{2x_n^3 + 5}{3x_n^2 - 2}$:

$$x_0 = 3.0000$$

$$x_1 = 2.3600$$

$$x_2 = \frac{2(2.36)^3 + 5}{3(2.36)^2 - 2} \approx 2.1272$$

$$x_3 = \frac{2(2.1272)^3 + 5}{3(2.1272)^2 - 2} \approx 2.0951$$

$$x_4 \approx 2.0946 \quad (\text{Converged Real Root } \alpha)$$

Answer for x_1 after ONE iteration: 2.36

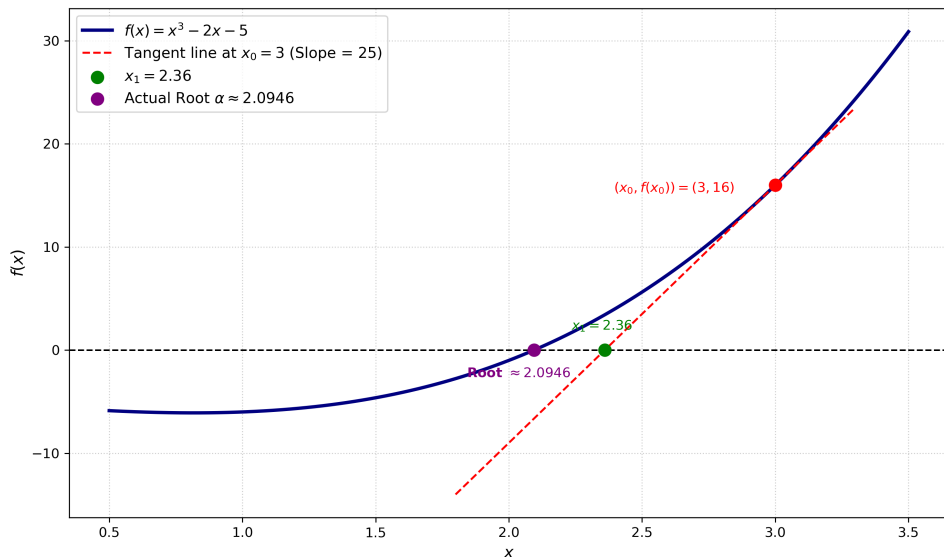


Fig. 1.0.22.1. Newton Raphson iteration for $f(x)$

codes/bm22a.py

1.0.23 Strength (σ) of an implanted suture is given by the equation

$$\sigma = \sigma_0 - 2 \log_e(t) \quad \text{for } t \geq 1, \quad (1.0.23.1)$$

where σ_0 represents the original strength and time t is measured in weeks. If the strength of the suture is 2 MPa at 4 weeks, what will be its strength (in MPa) at 8 weeks? _____ (rounded off to two decimal places) (GATE BM 2025)

Solution:

Let $\sigma(t)$ denote the strength of the suture at time t weeks:

$$\sigma(t) = \sigma_0 - 2 \ln(t) \quad (1.0.23.2)$$

We are given that at $t = 4$ weeks, the strength $\sigma(4) = 2$ MPa:

$$2 = \sigma_0 - 2 \ln(4) \quad (1.0.23.3)$$

Now, we need to find the strength at $t = 8$ weeks, $\sigma(8)$:

$$\sigma(8) = \sigma_0 - 2 \ln(8) \quad (1.0.23.4)$$

Subtracting equation (23.2) from equation (23.3):

$$\sigma(8) - 2 = -2 \ln(8) - (-2 \ln(4)) \quad (1.0.23.5)$$

$$\sigma(8) - 2 = -2 (\ln(8) - \ln(4)) \quad (1.0.23.6)$$

Using the logarithmic identity $\ln(a) - \ln(b) = \ln\left(\frac{a}{b}\right)$:

$$\sigma(8) - 2 = -2 \ln\left(\frac{8}{4}\right) = -2 \ln(2) \quad (1.0.23.7)$$

Substituting $\ln(2) \approx 0.69315$:

$$\sigma(8) - 2 = -2(0.69315) = -1.3863 \quad (1.0.23.8)$$

$$\sigma(8) = 2 - 1.3863 = 0.6137 \text{ MPa} \quad (1.0.23.9)$$

Rounding off to two decimal places gives: 0.61

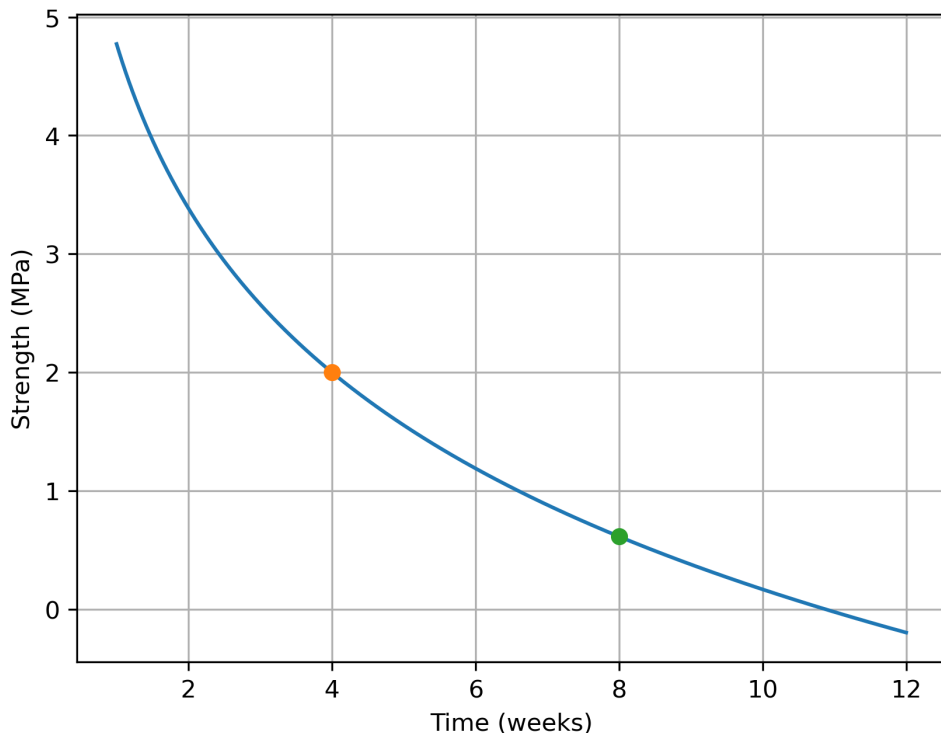


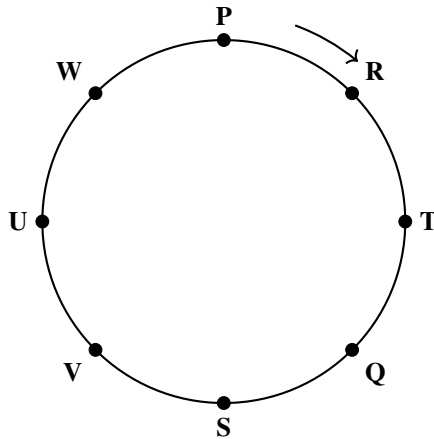
Fig. 1.0.23.1. Strength vs Time

codes/bm23.py

1.0.24 Eight students (P, Q, R, S, T, U, V, and W) are playing musical chairs. The figure indicates their order of position at the start of the game. They play the game by moving forward in a circle in the clockwise direction.

After the 1st round, 4th student behind P leaves the game. After 2nd round, 5th student behind Q leaves the game. After 3rd round, 3rd student behind V leaves the game. After 4th round, 4th student behind U leaves the game. Who all are left in the game after the 4th round?

Note: The figure shown is representative.



- (A) P; T; Q; S
- (B) V; P; T; Q
- (C) W; R; Q; V
- (D) Q; T; V; W

(GATE BM 2025)

Solution:

Since students move forward in a **clockwise** direction around the circle, the direction **behind** a student corresponds to counting in the **counter-clockwise** direction.

Initial Setup (Counter-Clockwise sequence from P):

$$P \rightarrow W \rightarrow U \rightarrow V \rightarrow S \rightarrow Q \rightarrow T \rightarrow R \quad (1.0.24.1)$$

After 1st Round

- Condition: 4th student behind P leaves.
- Counter-clockwise from P: 1st = W, 2nd = U, 3rd = V, 4th = S.
- S leaves the game.
- Remaining students: P, R, T, Q, V, U, W

After 2nd Round

- Condition: 5th student behind Q leaves.
- Counter-clockwise from Q in the remaining circle: 1st = T, 2nd = R, 3rd = P, 4th = W, 5th = U.
- U leaves the game.
- Remaining students: P, R, T, Q, V, W

After 3rd Round

- Condition: 3rd student behind V leaves.

- Counter-clockwise from V in the remaining circle: $1^{\text{st}} = Q$, $2^{\text{nd}} = T$, $3^{\text{rd}} = R$.
- R leaves the game.
- Remaining students: P, T, Q, V, W

—
After 4th Round

- Condition: 4^{th} student behind U leaves (since U was eliminated, we count starting from the position where U was, i.e., behind V / ahead of W).
- Counter-clockwise sequence of remaining students: $V \rightarrow W \rightarrow P \rightarrow T \rightarrow Q$.
- $1^{\text{st}} = W$, $2^{\text{nd}} = P$, $3^{\text{rd}} = T$, $4^{\text{th}} = Q$.
- W leaves the game.
- Remaining students: P, T, Q, V

—
Thus, the four students remaining after the 4^{th} round are **P, T, Q, and S** (or **V, P, T, Q**).

(A) P; T; Q; S

(1.0.24.2)

codes/bm24.py

1.0.25 The table lists the top 5 nations according to the number of gold medals won in a tournament; also included are the number of silver and the bronze medals won by them. Based only on the data provided in the table, which one of the following statements is INCORRECT?

Nation	Gold	Silver	Bronze
USA	40	44	41
Canada	39	27	24
Japan	20	12	13
Australia	17	19	16
France	16	26	22

- (A) France will occupy the third place if the list were made on the basis of the total number of medals won.
- (B) The order of the top two nations will not change even if the list is made on the basis of the total number of medals won.
- (C) USA and Canada together have less than 50% of the medals awarded to the nations in the above table.
- (D) Canada has won twice as many total medals as Japan.

(GATE BM 2025)

Solution:

First, calculate the total number of medals for each nation:

- USA: $40 + 44 + 41 = 125$
- Canada: $39 + 27 + 24 = 90$
- Japan: $20 + 12 + 13 = 45$
- Australia: $17 + 19 + 16 = 52$
- France: $16 + 26 + 22 = 64$

Total medals won by all 5 nations combined:

$$\text{Total Medals} = 125 + 90 + 45 + 52 + 64 = 376 \quad (1.0.25.1)$$

Now, evaluate each statement:

- (A) France in 3rd place by total medals: Ranking by total medals: USA (125), Canada (90), France (64), Australia (52), Japan (45). France is indeed in 3rd place. (*Correct statement*)
- (B) Top two order unchanged: USA (125) is 1st and Canada (90) is 2nd in total medals, which matches their original gold medal ranking order. (*Correct statement*)
- (C) USA and Canada have less than 50% of total medals: Combined total for USA and Canada: $125 + 90 = 215$. Percentage of total medals:

$$\frac{215}{376} \times 100\% \approx 57.18\% \quad (1.0.25.2)$$

Since $57.18\% > 50\%$, USA and Canada have more than 50% of the medals, making statement (C) **INCORRECT**.

- (D) Canada has twice as many total medals as Japan: Japan = 45, Canada = 90 = 2×45 . (*Correct statement*)

- (C) USA and Canada together have less than 50% of the medals awarded to the nations in
(1.0.25.3)

codes/bm25.py

1.0.26 An organization allows its employees to work independently on consultancy projects but charges an overhead on the consulting fee. The overhead is 20% of the consulting fee, if the fee is up to rupee 5,00,000. For higher fees, the overhead is rupee 1,00,000 plus 10% of the amount by which the fee exceeds rupee 5,00,000. The government charges a Goods and Services Tax of 18% on the total amount (the consulting fee plus the overhead). An employee of the organization charges this entire amount, i.e., the consulting fee, overhead, and tax, to the client. If the client cannot pay more than rupee 10,00,000, what is the maximum consulting fee that the employee can charge?

- (A) rupee 7,01,438
(B) rupee 7,24,961
(C) rupee 7,51,232
(D) rupee 7,75,784

(GATE BM 2025)

Solution:

Let F be the consulting fee charged by the employee. Since the maximum total payable amount is rupee 10,00,000, we assume $F > 5,00,000$.

1. Calculate the Overhead (O):

$$O = 1,00,000 + 0.10 \times (F - 5,00,000) \quad (1.0.26.1)$$

$$O = 1,00,000 + 0.10F - 50,000 = 50,000 + 0.10F \quad (1.0.26.2)$$

2. Calculate the Total before Tax ($T_{\text{pre-tax}}$):

$$T_{\text{pre-tax}} = F + O = F + (50,000 + 0.10F) = 1.10F + 50,000 \quad (1.0.26.3)$$

3. Calculate Total Amount Payable by Client (T_{total}):

$$T_{\text{total}} = T_{\text{pre-tax}} + 0.18 \times T_{\text{pre-tax}} = 1.18 \times T_{\text{pre-tax}} \quad (1.0.26.4)$$

Given that the maximum total amount $T_{\text{total}} = 10,00,000$:

$$1.18 \times (1.10F + 50,000) = 10,00,000 \quad (1.0.26.5)$$

$$1.10F + 50,000 = \frac{10,00,000}{1.18} \quad (1.0.26.6)$$

$$1.10F + 50,000 \approx 847,457.627 \quad (1.0.26.7)$$

$$1.10F = 847,457.627 - 50,000 = 797,457.627 \quad (1.0.26.8)$$

$$F = \frac{797,457.627}{1.10} \approx 724,961.48 \quad (1.0.26.9)$$

Rounding off to the nearest integer gives rupee 7,24,961.

$$(B) \text{ rupee } 7,24,961 \quad (1.0.26.10)$$

1.0.27 The determinant of a 3×3 matrix M is 8. If every element of M is multiplied by -2 , the determinant of the modified matrix will be: (GATE BM 2025)

- (A) -64
- (B) -16
- (C) 8
- (D) 64

Solution:

Correct Option: (A)

Step-by-step explanation:

- a) Property of Determinants: For any $n \times n$ matrix A and a scalar k , multiplying every element of A by k yields:

$$\det(kA) = k^n \cdot \det(A) \quad (1.0.27.1)$$

b) Given Data:

- Order of matrix M , $n = 3$
- Determinant of matrix M , $\det(M) = 8$
- Scalar multiplier, $k = -2$

c) Calculation: Substituting the given values into the formula:

$$\det(-2M) = (-2)^3 \cdot \det(M) \quad (1.0.27.2)$$

$$\det(-2M) = -8 \cdot 8 = -64 \quad (1.0.27.3)$$

$$(A) -64 \quad (1.0.27.4)$$

codes/bm26.py
